

Reg. No.: _____

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Amrita School of Computing, Bengaluru
B.Tech. Mid Semester Exam - March 2025
Computer Science and Engineering
Fourth Semester

23CSE214 Operating Systems

Time: Two hours

Maximum marks: 50

CO	Course Outcomes
CO1	Understand the architecture and functionalities of modern OS
CO2	Understand and apply the algorithms for resource management and scheduling
CO3	Analyze and Apply semaphores and monitors for classical and real-world synchronization scenarios
CO4	Engage in independent learning as a team and implement OS functionalities on a simulated machine.

Answer all questions

- 1
 - a Explain the system calls used to receive input from the keyboard and display output on the screen. [3M] CO1, BTL2
 - b Based on your program in (a) explain the sequence of states that the process moves from the start of its execution till its termination. [3M] CO1, BTL2
- 2 Differentiate between the following, with examples for each: CO1, BTL1
 - a Uniprocessor and Multi-processor [2M]
 - b Context switching and Process Swapping [2M]
 - c Single threaded and Multi threaded processes [2M]
- 3 With a neat diagram explain how system calls facilitate the transition from user mode to kernel mode. [4M] CO1, BTL2
- 4 A multi-threaded system runs about 60% parallel processes and the remaining of them run serially. Calculate the change in speed of the system if upgraded from a single core to a quad-core system. [2M] CO1, BTL2
- 5
 - a Given the below C Code: [2M+ 4M] CO1, BTL3


```
int main() {
    if (fork() == 0) {
        printf("Child process running\n");
    } else {
        wait(NULL);
        printf("Parent process\n");
    }
    return 0;
}
```

 - i) What is the purpose of wait(NULL); in the parent process? What happens if wait() is not used?
 - ii) Modify the given program so that the child process uses an external program to display its process ID.

- b. Consider two scenarios 2.5M CO1, BTL4
- i) Imagine a system that handles a series of critical financial operations, each processed in a separate thread. These operations involve complex calculations, such as interest computations or risk assessments, and updates to a shared database that tracks financial transactions. If any thread is interrupted midway through its operation, it could result in financial inconsistencies.

- ii) Imagine a system that periodically fetches weather data from an external API and displays it on a dashboard. Each data fetch operation is handled by a separate thread, and the system updates the display with the latest weather information. If a thread is interrupted midway, the system can simply retry the fetch operation in the next cycle, and the user might briefly see outdated data. 2.5M

Identify the thread cancellation strategy applicable for these scenarios, Justify your answer?

- 6 What is Inter-Process Communication (IPC)? Discuss different mechanisms used for communication between processes. 7M CO1, BTL2
7. Consider the following set of processes as shown in Table 1 with their arrival times, burst times, and priorities: CO2, BTL3

Table 1-Process Table

Process	Arrival Time	Burst Time	Priority
P1	2	5	3
P2	0	3	1
P3	1	9	4
P4	4	6	2
P5	3	2	5

- a) Draw the Gantt charts for the execution of these processes using the following CPU scheduling algorithms. Calculate Average Turn Around Time and Average Waiting Time. 3M
- i) Preemptive Priority Scheduling (higher number represents higher priority) 3M
- ii) Round Robin (RR) with Time Quantum = 3

- 8 a Describe the four scenarios that trigger a CPU scheduling decision and explain how each scenario determines whether preemptive or non-preemptive scheduling is applied. 6M CO2, BTL2
- b What is CPU- I/O Burst Cycle ? 2M CO2, BTL1

Course Outcome / Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO1	36	BTL 1	8
CO2	14	BTL 2	25
CO3	0	BTL 3	12
CO4	0	BTL 4	5
		BTL 5	0
		BTL 6	0